Gourav Singh

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Profile Summary

Aspiring Software Engineer with a strong academic record (CGPA: 9.29/10) and a foundation in object-oriented programming across Java, Python and C++, applied hands-on in Python projects. Familiar with data structures, algorithms and distributed systems through project work in backend services and performance monitoring. Proven problem-solving ability and time management through hackathons, **including Finalist at Goldman Sachs India Hackathon 2026 and 2nd place at AMD Slingshot**.

Education

Kalinga Institute of Industrial Technology — Bhubaneswar, India
CGPA: 9.29/10 | CSE (Third Yr.) — *August 2024 – Present*

Lakshmipat Singhania Academy — Kolkata, India
Score: 93% | Class XII (CBSE) | Science Club President — *June 2022 – Feb 2024*

B D Memorial International — Kolkata, India
Score: 96% | Class X (CBSE) | House Prefect — *March 2022*

Projects & Experience

ML-Based Credit Underwriting Risk Model

- Built a credit underwriting model that uses customer and banking data to predict loan default risk, achieving around **70–75% accuracy with precision and recall falling within 68–75%**, while explaining key risk drivers through **SHAP** and an interactive dashboard.
- Containerized the Streamlit application and ML inference pipeline using **Docker** and deployed it on **AWS EC2**, supporting **15–25 concurrent users** with **sub-2 second inference latency** for single-customer risk scoring.
- **Tech Stack:** Python, Pandas, NumPy, scikit-learn, SHAP, Matplotlib, Streamlit, Joblib, Docker, AWS EC2

Distributed AI Inference & Job Scheduling Platform

- Designed a **distributed AI inference system** using **FastAPI, REST APIs**, **Redis queue** and **3 background worker services**, structured with **object-oriented design** to process ML prediction jobs asynchronously under concurrent workloads.
- Implemented **concurrency, multithreading and synchronization** with **4 threads per worker**, enabling **12 concurrent worker threads**, job-state tracking and up to **3 retry attempts** for failed jobs.
- Simulated and tested the system with **1,000+ sample prediction jobs** to study **reliability, availability, throughput and latency**, improving operational efficiency through PostgreSQL-backed monitoring of failures and retries.
- **Tech Stack:** Python, FastAPI, Redis, PostgreSQL, Multithreading, Distributed Systems, REST APIs, scikit-learn, Streamlit, Docker, AWS EC2

Honors & Awards

Finalist, Goldman Sachs India Hackathon 2026 selected from 16,000+ participants, presenting the solution to a panel of VPs and senior engineers and iterating on feedback during mentorship sessions.
2nd Place, AMD Slingshot Regional AI/ML Ideathon organized by AMD and Hack2skill, competing against 100+ teams.
Winner, College Hackathon — awarded a cash prize of INR 10,000 for best technical solution.
Awarded by Taaza TV and School for excellent academic performance in Class XII and Class X.
Abacus Grandmaster & District Champion in mental arithmetic competitions.

Technical Skills

Languages: Java, Python, C++ (Object-Oriented Programming)
CS Fundamentals: Data Structures and Algorithms, OOP, DBMS
Systems/Concurrency: Distributed Systems, Multithreading, Synchronization, Deadlocks
Web/Backend: REST APIs, FastAPI
Databases/Auth: SQL, PostgreSQL
Data Science/ML: Machine Learning, Pandas, NumPy, EDA, Feature Engineering, Scikit-learn, Regression, Classification
Tools/Deployment: Git, GitHub, Docker, Power BI, Vercel, Render, Cloud Basics